Module 1: The Sun and Solar Wind

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1 Overview

In this module, we will talk about the sun: its different regions and temperatures, components of its radiation, its magnetic field, and more. The sun is the driving source behind tropospheric weather, of course, but also space weather. The interactions between the sun's radiation and its magnetic field with the near-Earth space environment drive all of the major space weather effects that we observe.

2 Key Points

- The sun is approximately a blackbody with a temperature of 5770 K.
- The visible wavelengths of the sun originate from the solar disk (the photosphere).
- The UV wavelengths of the Sun originate primarily from the chromosphere and corona.
- The sun's magnetic field is generated by its rotation; at the Earth's distance from the sun, this interplanetary magnetic field (IMF) is about 5-10 nanoTesla (nT).
- The sun's UV, X-ray, and particle outputs are highly variable over timescales ranging from minutes to decades.

Reading:

- Required: Pisacane, Chapter 5 (important), bits of Chapter 16
- Optional: Pisacane, Chapter 4 (just for fun)
- Optional: Tribble, Section 2.3.3.1 (pages 51–56)
- Optional: Tribble, Sections 2.2.2 (p. 31) & 2.3.2 (p. 49–51)
- Optional: Tascione, Sections 10.6–10.8 (p. 138–141)

3 Theory

The sun is categorized as a G-type main-sequence (G2V) star. Like all stars, it is a hot ball of plasma (i.e. ionized gas) with many regions, layers, and temperatures. Where we observe the surface of the sun, it has a radius of about 695,700 km (about 109 Earth radii) and a temperature of about 5700 Kelvins. This temperature determines the spectrum of light emitted by the sun, with its peak around 530 nm, i.e. in the yellow-orange part of the visible spectrum. The majority of the emitted light falls between 400 and 700 nm; this is also, conveniently, the wavelength range to which the human eye is sensitive. Coincidence?

In addition to the visible spectrum, however, the sun emits a large flux of radiation in the infrared (IR) and ultraviolet (UV) parts of the spectrum; a significant flux of X-rays; as well as ionized particles, primarily electrons (β particles), protons, and helium nuclei (α particles). These photons and particles originate from different regions of the sun's atmosphere. Compared to the visible solar spectrum, the UV, X-ray, and particle fluxes have enormous variability over time, and are responsible for the dynamic space weather effects that we observe in the near-Earth space environment.

3.1 Blackbody Radiation

Blackbody continuum radiation is created by constant random motion of electromagnetic oscillators in relatively dense, opaque matter. Blackbody radiators are equally good absorbers and emitters. Blackbody radiation from the Sun is in the UV to IR wavelengths. The temperature of the blackbody determines the amount of radiant power (photons/s) emitted at any given wavelength. By *Plank's Radiation Law*, which relates radiance of the body in a given wavelength interval to its temperature,

$$R(\lambda, T)d\lambda = \frac{2\pi hc^2}{\lambda^5}\left(\frac{1}{e^{(hc/\lambda k_B T)} - 1}\right)d\lambda \tag{1}$$

$$= \frac{c_1}{\lambda^5}\left(\frac{1}{e^{(c_2/\lambda T)} - 1}\right)d\lambda \tag{2}$$

where

$$\begin{aligned}
R &= \text{radiated energy flux per wavelength [W/(m}^2\text{ nm)]} \\
\lambda &= \text{wavelength [m]} \\
T &= \text{temperature [T]} \\
h &= \text{Plank's constant}(6.62 \times 10^{-34}\text{J s}) \\
c &= \text{light speed}(2.998 \times 10^{8}\text{m/s}) \\
c_1 &= 3.74 \times 10^{-16}\text{W m}^2 \\
c_2 &= 1.44 \times 10^{-2}\text{m K.}
\end{aligned}$$

Differentiating the Plank function produces *Wien's Displacement Law*:

$$\lambda_{max} T = \alpha \tag{3}$$

where

$$\begin{aligned}
\lambda_{max} &= \text{wavelength for maximum amount of radiation [m]} \\
T &= \text{object temperature [T]} \\
\alpha &= \text{constant}(2.898 \times 10^{-3}\text{m K}).
\end{aligned}$$

Integrating the Plank equation over all wavelengths produces the total radiance, $R(T)$, emitted to space from a unit area of the body, called the *Stefan-Boltzmann Law*:

$$R(T) = \int_0^{\text{inf}} R(\lambda, T)d\lambda \tag{4}$$

$$= \varepsilon_\lambda \sigma T^4 \tag{5}$$

where

$$\begin{aligned}
R(T) &= \text{energy flux (radiance) over all wavelengths [W/m}^2\text{]} \\
\varepsilon_\lambda &= \text{emissivity of radiant efficiency [unitless]} \\
\sigma &= \text{Stefan-Boltzmann Constant}(5.67 \times 10^{-8}\text{W / (m}^2\text{ K}^4)).
\end{aligned}$$

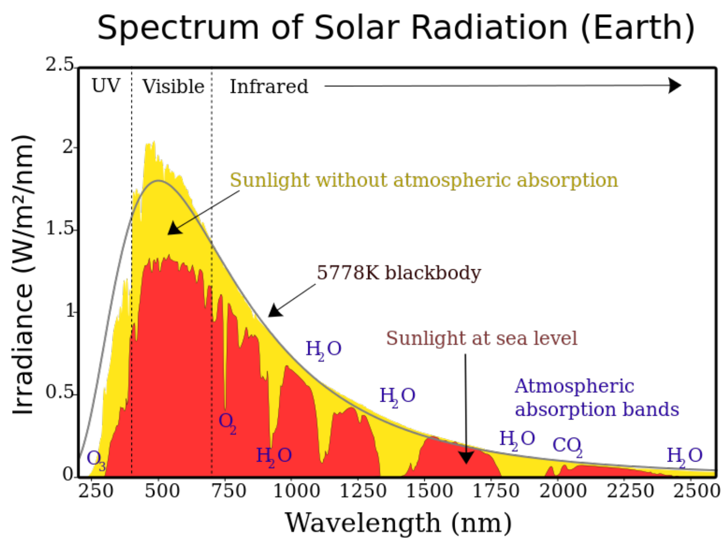

Figure 1: Solar spectrum in yellow compared to a 5778 K blackbody in black with atmospheric absorption effects shown in red [source link] .

For all blackbodies, ε equals one. When a blackbody radiates uniformly from its entire surface, the total energy flux, or luminosity, is given by:

$$L = \varepsilon\sigma AT^4 \tag{6}$$

where

$$L = \text{luminosity [W]}$$
$$A = \text{radiating area [m}^2\text{]}.$$

3.2 Discrete Line Radiation and Absorption

Electrons in atoms can only possess discrete amounts of energy, and when they move between these energy levels, they emit or absorb energy as specific frequencies. The lowest energy level is called the ground state, and higher levels are called excited states. A low density gas will create line spectra when electrons transition between bound energy states or between a free and bound state. Specifically, the H–α (primary hydrogen Balmer) transition from n=3 to n=2, which produces radiation at 656.3 nm, is used for monitoring the sun. Solar H–α comes from the photosphere and chromosphere.

3.3 Solar Interior

The solar interior is described by three major regions, the *core*, the *radiative zone*, and the *convective zone*. The boundary between the radiative and convective zone is called the *tachocline*. The regions are differentiated mainly by energy transport method. Figure 2 depicts the internal structure of the sun, labeling the core, radiative zone, and convection zone.

3.3.1 Core

The core of the sun spans from $0R_s$ to $0.25R_s$ and the temperature is about 1.5×10^7 K. The pressure is approximately 2×10^{16} Pa. The core is so hot and dense that hydrogen fuses into helium. This process of

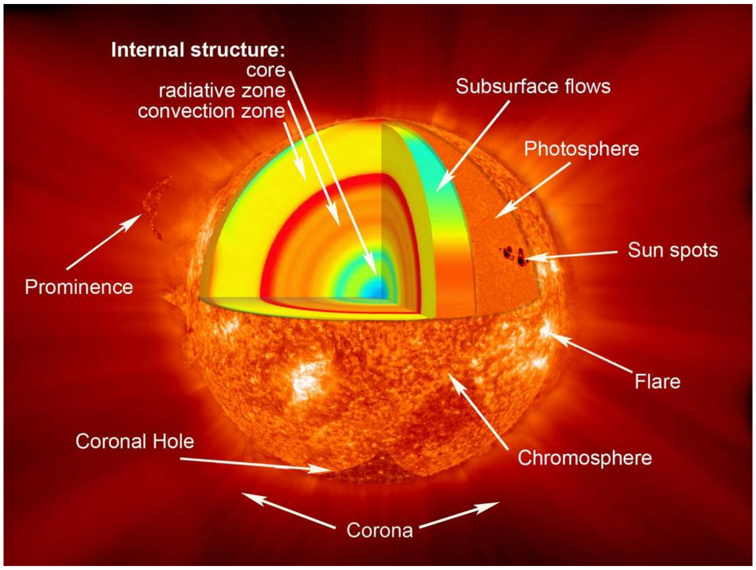

Figure 2: Illustration of the regions of the sun and some smaller features labeled in white [source link].

hydrogen converted to helium in the sun is called the *proton-proton (PP) chain* and is depicted in Figure 3. The PP chain generates the Sun's energy, converting four hydrogen nuclei to one helium nucleus and releasing 26.73 MeV of energy with each reaction. This energy is divided between 26.23 MeV gamma rays and 0.5 MeV neutrinos. The photons (gamma rays) released remain trapped in the sun for millions of years as they slowly diffuse outwards in a "random walk," much longer than the 8 minutes the photon will travel from the surface of the sun to the Earth.

3.3.2 Radiative zone

After leaving the core, photons continue their journey outward and enter into the radiative zone. The radiative (or radiation) zone spans from $0.25R_s$ to $0.7R_s$. The temperature in this region decreases with increasing radial distance. Indicated by the name, the main method of energy transfer is *radiative diffusion*. Due to the high density of this region, photons can only travel a short distances without interacting with other particles. With this repeated absorption and scattering, photon paths resemble the zigzagging white lines in Figure 4, and can hundreds of thousands of years to travel through the radiative zone. This is the "random walk" mentioned above.

3.3.3 Tachocline

The tachocline is the boundary between the radiative and convective zones and is characterized by the transition in energy transport method from radiative diffusion to *convection*. This boundary is about 0.1 R_s in width and is located at about 0.7 R_s. It has a temperature of $\geq 5 \times 10^6$ K. This boundary interfaces the solid body rotation of the lower layers with the differential rotation above. Due to these changing flow velocities, this is likely where the Sun generates, stores, and amplifies magnetic fields.

Sheared flow deforms polodial magnetic fields into azimuthally stretched torodial magnetic fields. Concentrating the magnetic field, this creates high pressure zones that in turn push out plasma to create low mass density bubbles. These bubbles are buoyant and move upwards, and can push up the concentrated fields, creating the active regions that drive space weather. This cycling is called the *solar dynamo*. The solar dynamo generates the solar magnetic field by converting kinetic energy (through plasma convection, meridional circulation, and differential rotation) into electromagnetic energy.

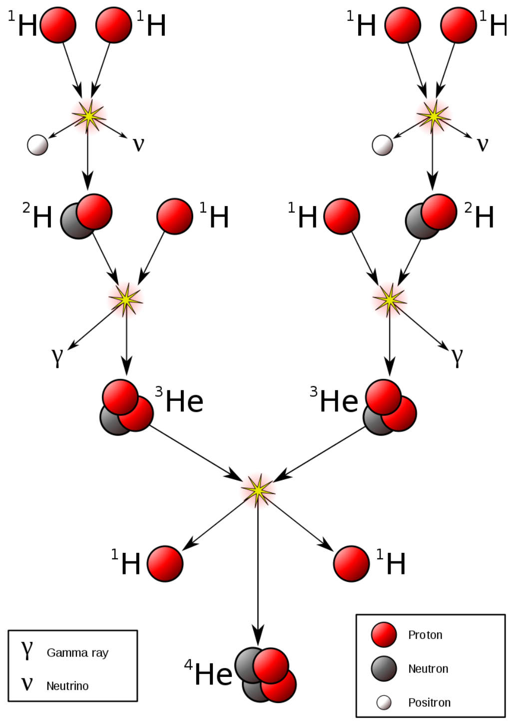

Figure 3: PP chain illustration showing the process of converting hydrogen to helium that occurs in the solar core [source link].

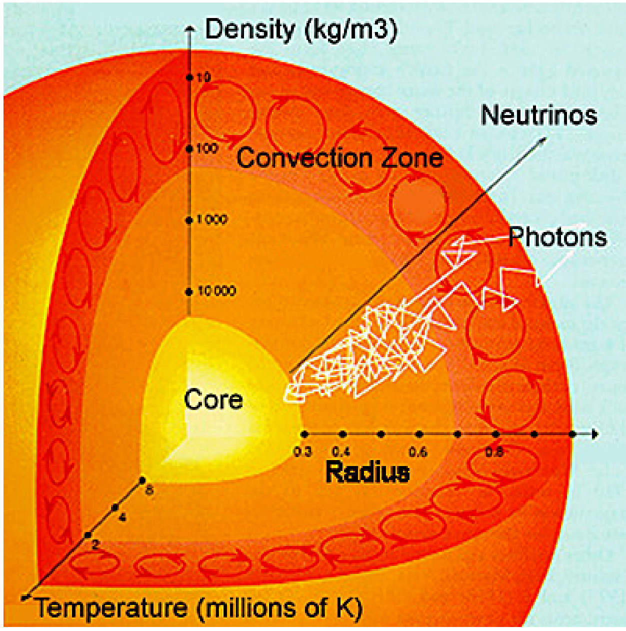

Figure 4: Illustration of the solar interior showing the "random walk" of photons through the radiative zone in white.

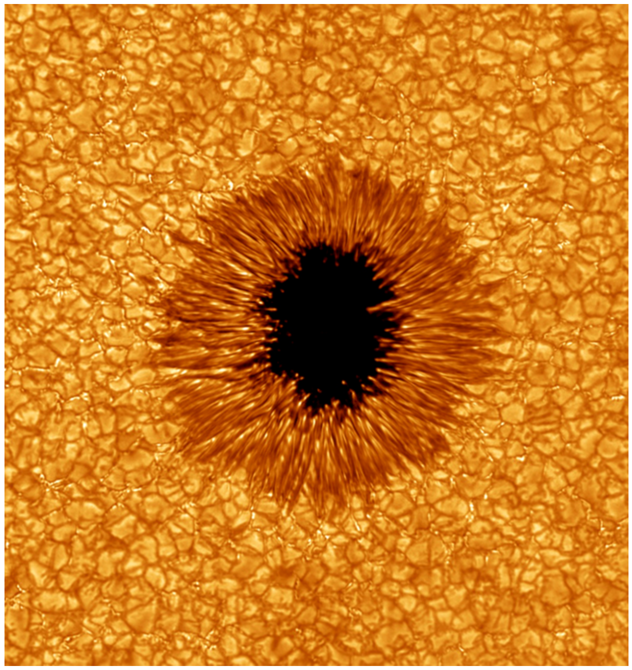

Figure 5: The dark spot in the center of this image is a sunspot. The threadlike structures surrounding the sunspot are called faculae. The bubbling structures covering the rest of the image are called granules, which span about 1000 km each.

3.3.4 Convective zone

The convective zone spans from $0.7R_s$ to $1.0R_s$, and transfers energy through convective fluid motion, analogous to boiling water. In a simplified explanation, plasma is heated and rises then cools and descends as newly heated plasma rises. The flow pattern this creates is depicted in Figure 4 as red circular arrows.

The convective motion is organized into three scales of granulation, listed from smallest to largest: *granules*, *supergranules*, and *giant convective cells*. Granules, shown in Figure 5 surrounding a sunspot, cover the surface level of convection and are about 1000 km across. They last an average of 20 minutes, and they recycle magnetic field. The larger, deep convection supergranules (Figure 6) span an average of 30,000 km and last for 1 to 2 days. They produce the roots and wave structure in solar wind, as well as concentrate B field. Finally, giant convection cells (Figure 7) are about 100 x 500 Mm and last about six months. They may support differential rotation and “active longitudes” or repeating active regions.

The convective zone is also characterized by *differential rotation* (Knipp p. 95). The solar equatorial region rotates faster than the polar regions. With this large scale shearing motion, angular momentum is redistributed and turbulence is created. The effects of differential rotation on the solar magnetic field is shown in Figure 8.

3.4 Solar Atmosphere

The solar atmosphere refers to the regions of the sun from the surface outward, including the *photosphere*, *chromosphere*, *transition region*, and *corona*. These regions are where most source signatures of space weather events are observed, from *solar flares* that accelerate high energy particles towards the Earth to the *coronal hole* sources of *high speed streams* of solar wind. Figure 2 labels these regions, as well as some some smaller features. We also include the *heliosphere* in this section, which refers to the entire region within the influence of the sun.

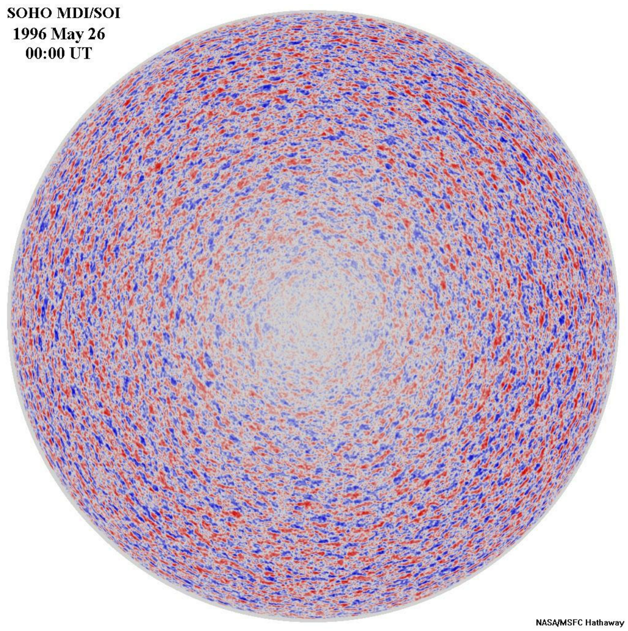

Figure 6: Convection supergranules which each span an average of 30,000 km.

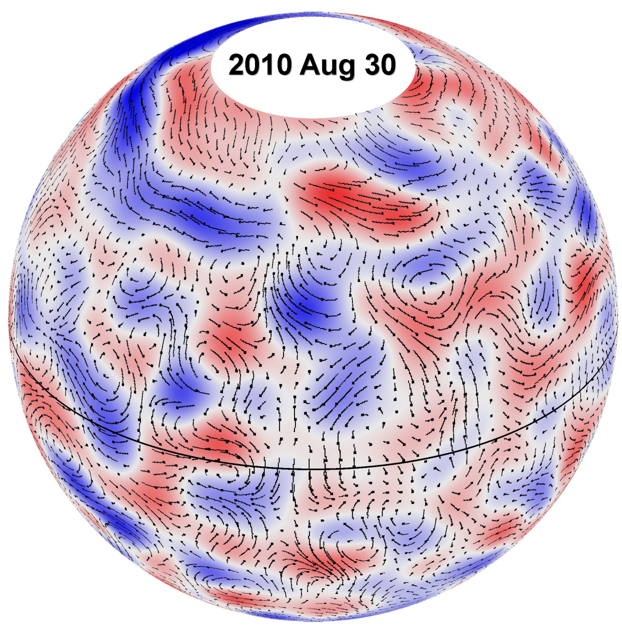

Figure 7: Giant convection cells, which are about 100 x 500 Mm.

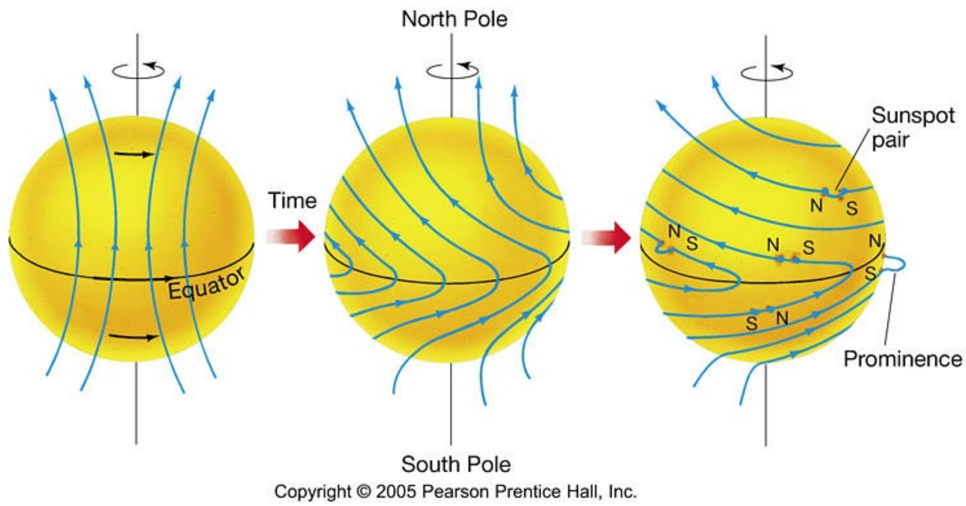

Figure 8: The differential rotation of the sun deforms the solar magnetic field giving rise to prominences (filaments off the solar limb) and active regions.

3.4.1 Photosphere

The photosphere is referred to as the "surface" of the sun, or the boundary where visible light photons have a good chance of escaping the sun. The photosphere is less than 500 km thick with a very low density $\approx 10^{-4}$ kg/m^3. The photosphere is the source of 99% of visible and IR radiation at Earth, and is considered a black body radiator. The temperature is relatively cool, $\approx$5800 K, and decreases with height.

Sunspots are observable on the photosphere, appearing dark compared to the surrounding area in visible light (Figure 5). Sunspots can be thought of as the "footprints" of loops of intense magnetic field called active regions, as they are located where the loops pass through the photosphere.

3.4.2 Chromosphere and Transition Region

The chromosphere starts where the photosphere temperature reaches a minimum and is about 2 Mm thick. The temperature increases from about 4,300 K to 20,000 K. The chromosphere has a low density and is transparent at visible wavelengths, contributing little to radiative output. However it is an effective radiator of line emissions, particularly the *Hydrogen-Balmer line* and the *Ca-II line*. The Hydrogen-Balmer-α (H-α) line is the electronic transition in the resident neutral hydrogen atoms at 656.3 nm, a transition from n=3 to n=2. Figure 11 shows the chromosphere imaged in H-α on the left compared to the photosphere in visible light on the right. Ca-II, radiation from singly ionized calcium, is used to visualize the magnetic field. Figure 12 shows the chromosphere imaged in Ca-II.

The transition region is a thin layer (100 km - 200 km) above the chromosphere where the temperature rises sharply from 10^4 K – 10^6 K. This is where filaments (called prominences when viewed off the solar limb instead of on the solar disk) reside. Filaments are relatively cool (7000 K – 15,000 K compared to 10^5 K – 10^6 K), dark, dense, thread-like structures suspended in the upper atmosphere. They typically have long lifetimes on the sun, but can "peel" off and escape the sun as coronal mass ejections (CMEs), especially when underlying plasma is disturbed.

3.4.3 Corona

The corona is the hottest region of the outer solar atmosphere (1 MK – 2 MK in magnetically quiet regions, 2 MK – 5 MK in magnetically active regions, hotter in flaring regions), but so thin and tenuous that you

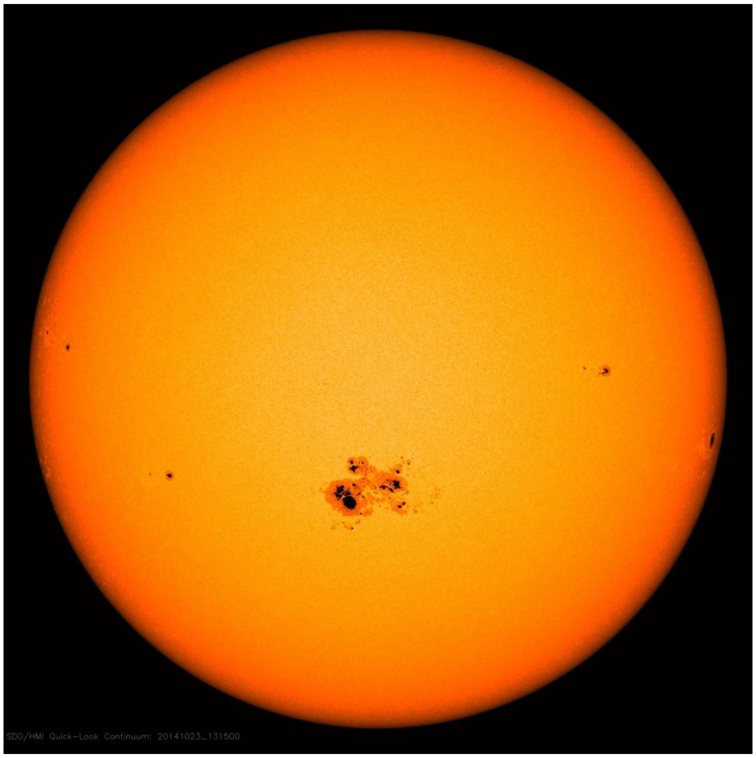

Figure 9: Visible light image of the sun showing several sunspot regions, the largest group just south of center disk [source link].

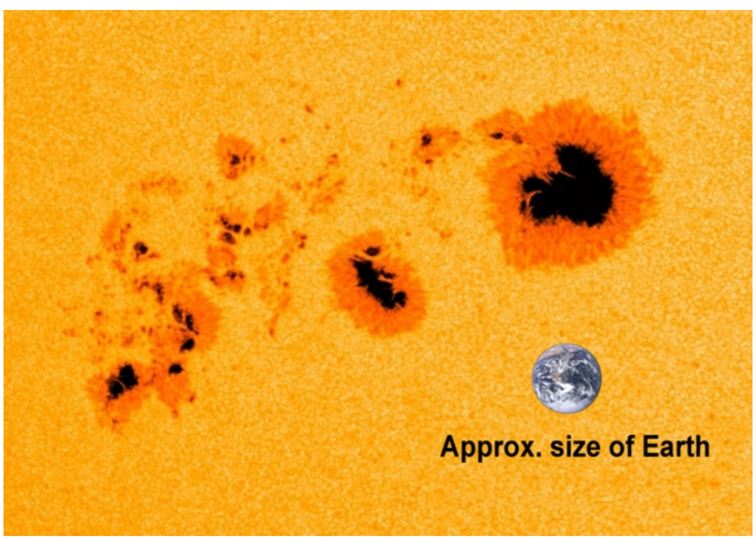

Figure 10: A closer look at sunspots with the size of the Earth shown for scale [source link].

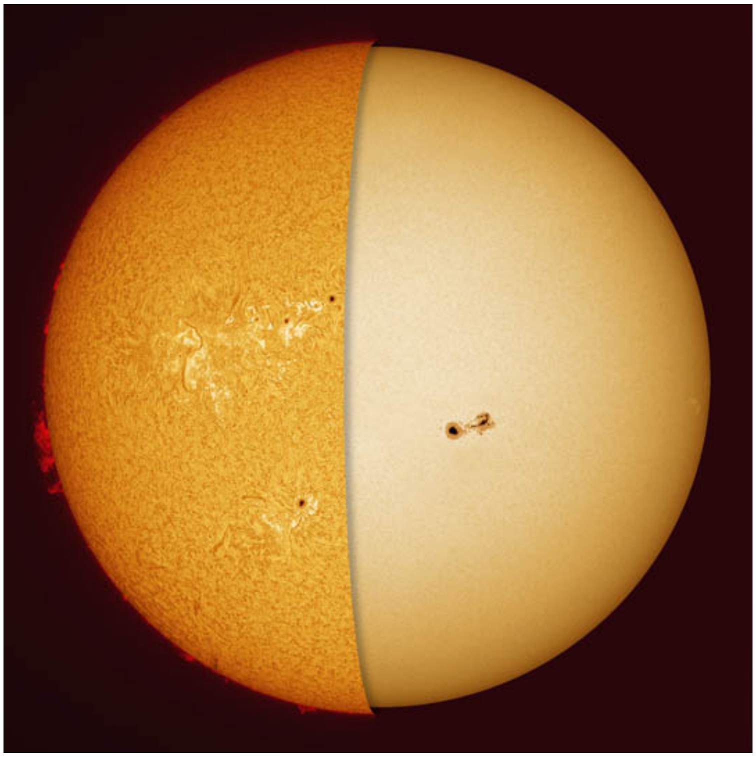

Figure 11: Chromosphere in H-α (left) and photosphere in visible light (right).

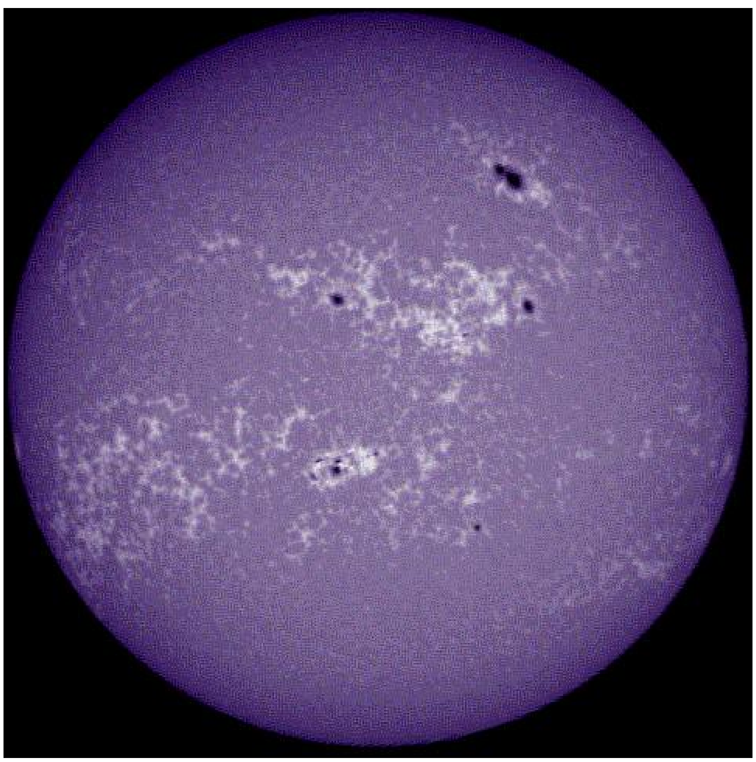

Figure 12: Chromosphere imaged in Ca-II.

would feel cold! The corona is illuminated by white photospheric light scattering of coronal electrons. However, this is so dim compared to the photosphere that the corona can not be view without an eclipse to block the solar surface. *Coronagraphs*, instruments that image the corona, create artificial eclipses for their detectors by using occulting disks to block the solar surface. Space weather forecasters actively use data from the coronagraphs (C1 and C2) onboard the Solar and Heliospheric Observatory (SOHO), as well as on STEREO–A (lost communication with STEREO–B) to measure coronal mass ejections (CMEs).

The corona can be treated as a quasi-blackbody with strong emission lines. The ionizing radiation from the corona is responsible for the Earth's ionosphere, as well as the ionospheres of other planets. Escaping coronal plasma creates the solar wind.

3.4.4 Heliosphere

The outermost portion of the solar atmosphere is known as the heliosphere. This region, which encompasses all of the planets in our solar system, consists of the solar wind, the constant flow of plasma from the sun. The flow of plasma includes a frozen in magnetic field, the interplanetary magnetic field (IMF). These field lines follow a spiral pattern due to the rotation of the sun called the Parker spiral. Visualized in Knipp Fig. 5-15 (p. 218), you can also imagine this phenomena by picturing someone holding a water hose while spinning. The water from the hose would create a spiral shape due to the person's rotation. How the solar wind and IMF vary with solar activity is discussed in Section 3.5. The heliosphere extends to the termination shock somewhere between 75 and 90 AU and was observed by Voyager 1 and Voyager 2. Beyond the termination shock is the heliosheath and heliopause. You can estimate the heliopause location as where the solar wind pressure becomes equal that of the interstellar medium.

3.5 Solar Activity

All solar activity is driven by magnetic processes in the Sun. Regions where magnetic field lines are open, called coronal holes, create high speed streams of solar wind. Magnetic field lines twist together and concentrate magnetic flux necessitating a release of magnetic energy in the form of a flare and/or a coronal mass ejection. Flares accelerate solar energetic particles at relativistic speeds away from the sun. Coronal mass ejections (CMEs), billion-ton plasma eruptions, plow through the heliosphere. This section covers all of the above phenomena, as well as the large scale pattern of activity called the solar cycle.

3.5.1 Solar Magnetic Field and Solar Cycle

The solar cycle is an 11 year periodic change in solar activity where the solar activity increases to solar maximum and then decreases to return to solar minimum. You may hear that a solar cycle is actually 22 years. This is because the polarity of the magnetic field flips every 11 years, so some define a full cycle of solar activity as 22 years to account for this. A cycle of solar activity was first noticed in sunspot number records (Figure 13. The sunspot record is the longest record of solar activity, as the number of sunpots correlates to the number of active regions, which increases as solar activity increases. We are currently in past the maximum of Solar Cycle 25 and just past the solar maximum, entering the declining phase.

On a global scale, the solar magnetic field generally has a tilted dipole structure. At the surface, the intensity is about 10^{-4} T, which is 2-3 times stronger than Earth's magnetic field. The polar region initially has open field lines, while the equatorial region has a closed field. As the sun progresses through its solar cycle, this polar concentration of open field shifts towards the equator, with more coronal holes found in equatorial regions halfway through the solar cycle. Some common features observed on the Sun that indicate high magnetic flux are faculae, plages, spicules, active regions, and prominences, all defined in Tascione 2.3. Active regions are areas with clusters of sunspots and high magnetic flux where most flares and CMEs originate.

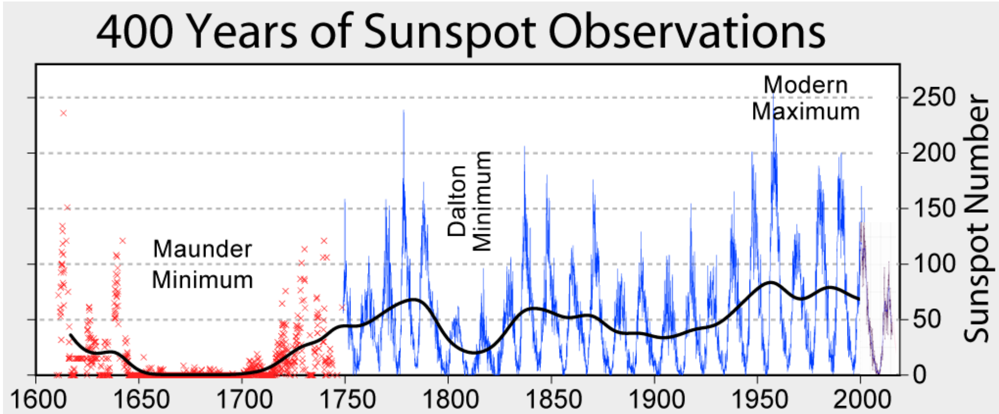

Figure 13: Sunspot number record showing correllation to the 11 year solar cycle.

3.5.2 Solar Flares

Solar flares are huge bursts of EM radiation, mainly in X-ray and EUV wavelengths, from the Sun that accelerate solar energetic particles (SEPs) to relativistic speeds. These particles travel along magnetic field lines, and when these lines are well connected to the Earth, the SEPs can have damaging effects on spacecraft near Earth. Flare effects can even block radio communications! This section will give a brief overview of flares, with more detail provided in Tascione 2.4 and 2.5.

Flares typically originate in active regions, the areas on the sun with clustered sunspots and high magnetic intensity. To understand why a flare occurs, imagine the looping field lines of active regions described earlier in Section 3.4.1. As the region continues to build up magnetic energy and field lines continue to twist, the growing instability of the region necessitates that energy be released. As illustrated in Figure 14, magnetic reconnection is the process through which the energy is released and the flare is produced. This same process can also create a coronal mass ejection (CME). However, CMEs and flares can also happen independently.

The large EM eruptions of solar flares can last from minutes to hours, and they are typically observed as bright flashes in EUV images of the Sun (Figure 15). An atmospheric response to the flare can be observed at almost the same time from the energy that is released traveling at the speed of light. Specifically, radiation from the flare increases ionization in the ionosphere's lower layers. This effect is what causes radio wave blackouts that will be covered more in the Ionosphere module later this semester. NOAA Space Weather Prediction Center (SWPC) classifies flares based on the likeliness of radio blackout and flare flux, as shown in Table 1

NOAA SWPC Solar Flare Classification

Radio Blackout	X-ray Flare	Flux (W/m2)	Severity Descriptor
R1	M1	0.00001	Minor
R2	M5	0.00005	Moderate
R3	X1	0.0001	Strong
R4	X10	0.001	Severe
R5	X20	0.002	Extreme

Table 1: NOAA SWPC's table showing the correlation between radio blackouts, solar flares, nominal energy flux (watts per square meter), and the designated severity event descriptor.

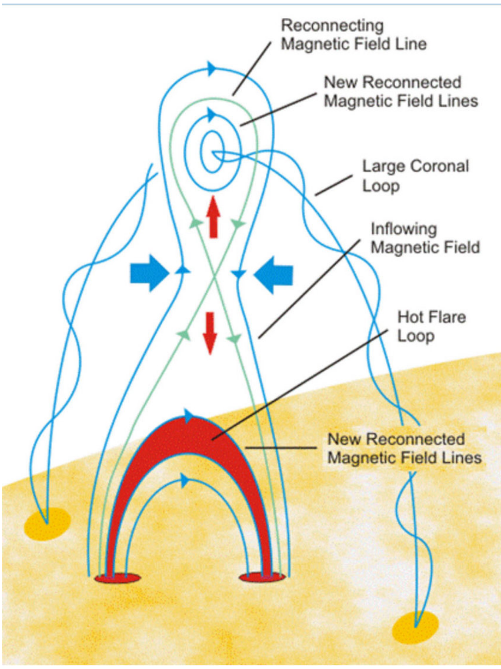

Figure 14: Magnetic reconnection in an active region.

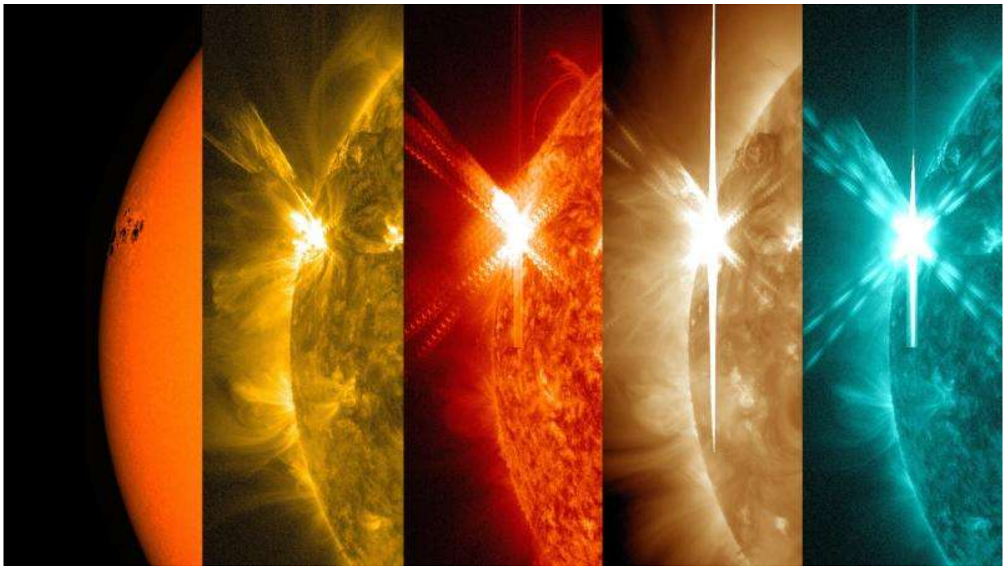

Figure 15: Solar flare observed by SDO from an active region with a cluster of sunspots in the following wavelengths (left to right): visible light, 171 Å, 304 Å, 193 Å, and 131 Å.

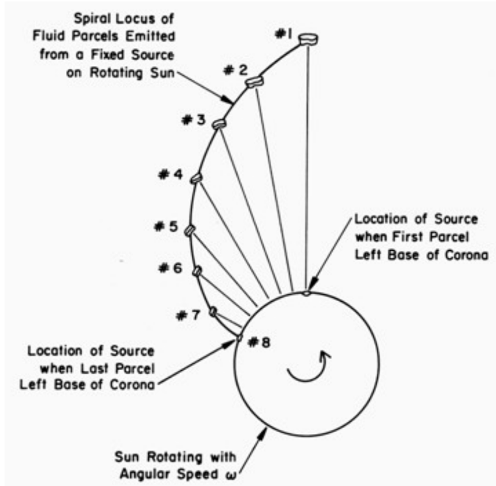

Figure 16: Illustration of how the Parker spiral is formed showing the progression of plasma particles emitted from the same point on the Sun.

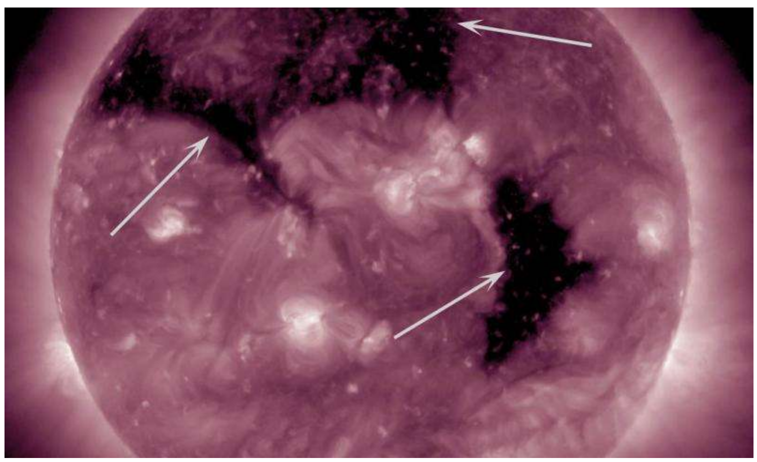

Figure 17: SDO image in 211 Å with white arrows pointing to the dark coronal hole regions.

3.5.3 Solar Wind

We began discussion of the solar wind in Section 3.4.4. Recall that the solar wind is the constant flow of plasma from the sun, which mainly consists of electrons and protons. The flow of plasma includes a frozen in magnetic field, the interplanetary magnetic field (IMF). These field lines follow a spiral pattern due to the rotation of the sun called the Parker spiral (Figure 16). The solar wind conditions can cause their own space weather effects at Earth. Additionally, solar wind, as the medium through which everything in the solar system travels, also affects how phenomena like CMEs travel and develop.

The speed of the solar wind depends on the source region on the Sun. Higher speed wind flows from regions called coronal holes, which appear dark in EUV imagery. Of course, these are not actual holes in the Sun. Coronal holes are regions with open magnetic field lines that allow for the faster solar wind speeds, ranging from about 500 to 800 km/s. A fast solar wind stream is called a high speed streams (HSS). High speed streams can cause geomagnetic storms at Earth, and a long-lasting HSS at Earth is often responsible for the radiation belt enhancements we will cover in the radiation belt module.

HSS regions also have a different density than the nominal solar wind, which travels closer to 400 km/s. The shear between the high and low speed regions creates a boundary region called the co-rotating

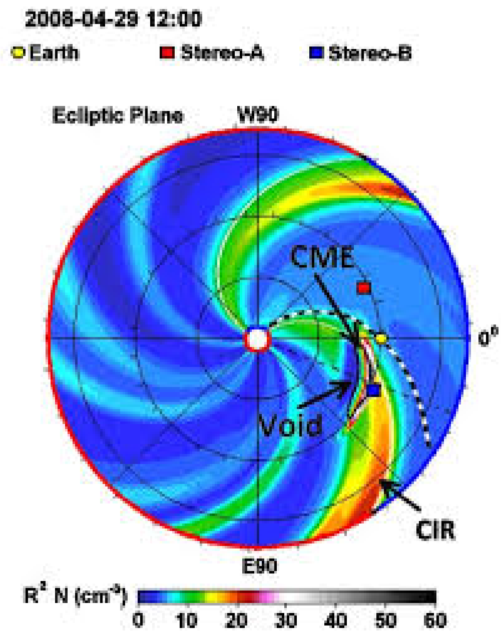

Figure 18: A WSA–ENLIL+Cone model of the solar wind density with the CIR and a CME labeled.

interaction region (CIR). CIRs have very high density and strong magnetic fields. Figure 18 shows a model of the solar wind, called the WSA–ENLIL+Cone model, with the CIR region pointed out. CIRs can create an interplanetary shock in the plasma that drives geomagnetic storms when interacting with the Earth's magnetic field. CIRs can be confused with coronal mass ejections (CMEs) in in-situ observations, but the distinctions will be discussed more in the next section.

Section 3.5.1 discussed that the distribution of coronal holes with latitude changes with the solar cycle. Coronal holes are centered on the poles during solar minimum, and they move to equatorial latitudes by solar maximum. Thus, the frequency of high speed streams with the potential to effect the environment near Earth and other bodies in the equatorial plane is higher during solar max.

Due to the polarity of the solar wind magnetic field, an equatorial current sheet is created with one B field polarity to the north of the sheet and the opposite to the south. At quiet times, this sheet is relatively flat, but with high activity the sheet becomes distorted to a "ballerina skirt" shape due to active regions and coronal holes invading the equatorial region (Figure 19). We can observe current sheet near Earth as we travel through it, with solar wind polarity alternating as we travel through the dips and peaks of the ballerina skirt current sheet.

3.5.4 Coronal Mass Ejections

Coronal mass ejections, or CMEs, are eruptions of billions of tons of plasma from the sun that plow through interplanetary space and cause geomagnetic storms when they interact with the Earth's magnetic field. CMEs are made of coronal plasma and carry the magnetic field of the plasma with them. This internal magnetic field is called the flux rope. For larger and faster CMEs, the flux rope has a stronger magnetic field than the surrounding solar wind. Speeds can range from 250 km/s to nearly 3000 km/s, but an average Earth-directed CME will take about three days to travel from the Sun to the Earth. Due to the complicated plasma physics of CMEs, how they interact with the solar wind, and how they can interact and combine with each other, CMEs are difficult to model for forecasting purposes.

CMEs have two major methods of source eruption: active region eruption and filament eruption. Active region eruptions are caused in much of the same way as solar flares. The magnetic flux of an active region

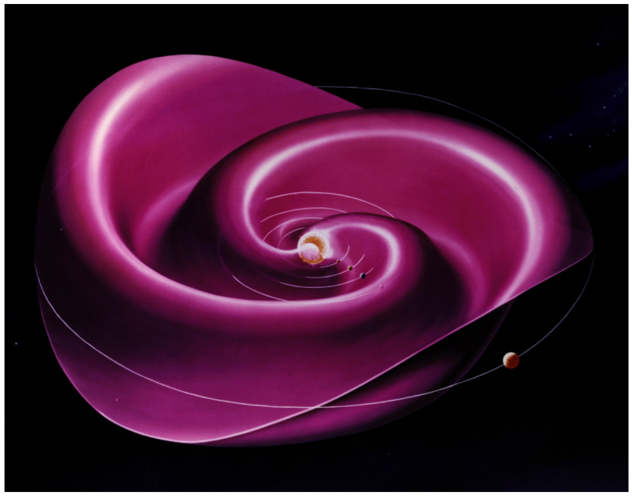

Figure 19: Depiction of the heliospheric current sheet showing the "ballerina skirt" shape.

Property	Fast Wind	Slow Wind	Average Wind
Proton Temperature	1.5×10^5 K	0.5×10^5 K	1.2×10^5 K
Electron Temperature	0.75×10^5 K	2×10^5 K	1.4×10^5 K
Bulk Speed	450–800 km/s	250–450 km/s	468 km/s
Speed Variation	5%	25–50%	15%
Sound Mach Number	11–20	6–11	11
Alfvén Mach Number	5–10	3–5	5
Ion number density	3 cm^{-3}	10 cm^{-3}	8.7 cm^{-3}
Composition	95% H, 4% He	94% H, 5% He	95% H, 4% He
Magnetic Field	$\sim$5 nT	$<$5 nT	$\sim$6.2 nT
Original at Sun	Coronal holes	Streamer boundaries	
% Occurrence	44%	34%	

Table 2: Solar wind flow parameters; from Knipp, Table 5-1.

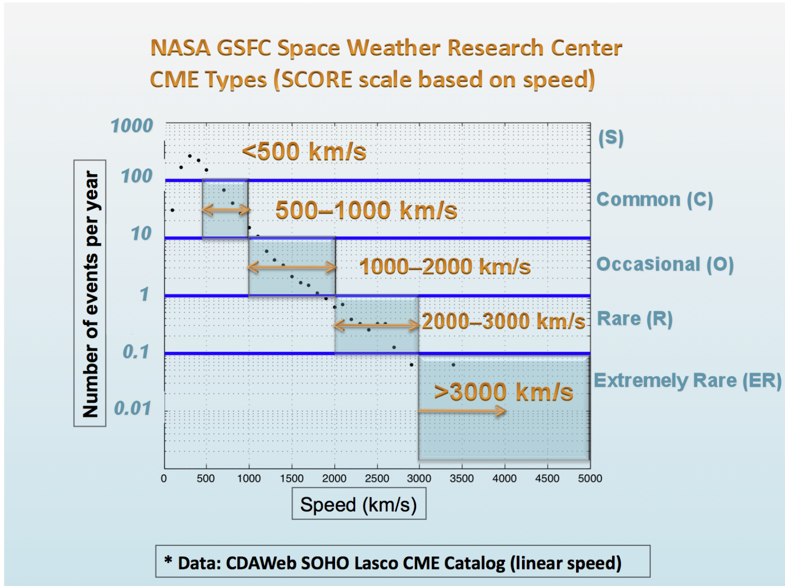

Figure 20: The CME SCORE classification indicates the type of the detected CME based (S-type, C-type,...). The type also indicates the rate of occurrence of each type of CME from most to least likely.

builds until a breaking point of energy release, magnetic reconnection, occurs. Figure 14 shows this process of magnetic reconnection over an active region. The magnetic reconnection is occuring in the center "X" region, and a CME occurs when the upper bubble of plasma is released and escapes the corona. Because this same process produces flares, we often see flares and CME source eruptions concurrently, but one does not necessitate the other. In SDO images of the Sun, eruptions from active regions can be identified from several non-flare signatures, including EUV waves, dimming, and bright post-eruptive arcades.

The second possible source for a CME is a filament. Recall from discussing the transition region that filaments are relatively cool, dark, dense, thread-like structures suspended in the upper atmosphere of the Sun. They typically have long lifetimes on the sun, but can "peel" off and escape the sun as coronal mass ejections (CMEs), particularly when underlying plasma is disturbed. Often flares are observed below a filament that triggers the filament to begin to erupt. If the filament plasma is not reabsorbed and escapes the corona, a CME is produced. If the filament is seen on the limb (edge) of the sun, it can be called a prominence.

CMEs are categorized according to speed, shown in Figure 20, but size and direction are also essential to predicting CME arrivals. Additionaly, CME density and magnetic field strength and orientation have a huge impact on how a CME will interact with the Earth's magnetic field. If a CME's flux rope has a B_z component that point southward (negative), the CME will interact more with the magnetosphere, which has northward (positive) B_z. These oppositely directed B_z components are ideal for magnetic reconnection and cause more intense geomagnetic storms, which we will cover in later modules. Ahead of fast CMEs, a shock wave called the sheath is formed. The sheath can accelerate charged particles and cause an SEP event at Earth that arrives slower than a flare-accelerated SEP event.

Material	Absorbance (α)	Emittance (ϵ)	Ratio α/ϵ
Aluminum	0.10	0.05	2.000
Black Paint	0.97	0.89	1.090
FEP* – 2 mil	0.06	0.68	0.088
FEP – 5 mil	0.11	0.80	0.138
Fused Silica	0.08	0.81	0.099
Gold	0.21	0.03	7.000
Grafoil	0.66	0.34	1.941
Kapton	0.48	0.81	0.593
Kapton, coated	0.40	0.71	0.563
Kapton, Au film	0.53	0.42	1.262
OSR*, coated	0.09	0.76	0.118

Table 3: Absorbance and Emittance of common spacecraft materials. *FEP = Fluorinated Ethylene Propylene, used as wire insulation. OSR = Optical Solar Reflector, a layer of quartz on a metal substrate. Taken from Tribble p. 53.

4 Effects

Most of the effects we are interested in are ultimately driven by the sun, simply because the space environment is driven by the sun. We'll cover most of these effects in later sections. Here, we'll focus on a few basic effects of the sun.

4.1 Thermal

A spacecraft exposed to solar radiation gets warm. Every material (solid, liquid, gas, and plasma) absorbs radiation to some degree, and this absorption depends on the wavelengths of that radiation. Here on the surface of the Earth, the absorption of radiation is balanced by a number of cooling mechanisms: radiative cooling (emittance of radiation such as infrared), convective cooling (as cool air pulls heat away), or conductive cooling (direct contact with a cooler material).

In space, there is no convective cooling. Spacecraft surfaces can conductively cool or heat each other, but that heat cannot leave the spacecraft conductively. The only way to get rid of excess heat is by radiation. The efficiency with which a material absorbs radiation[1] is called its *absorbance*, and the efficiency with which it emits radiation is its *emittance*. Typical values for these coefficients are given in Table 3.

The heat absorbed (Watts) by an object with absorbance α is given by

$$Q_{in} = \alpha A_n S \tag{7}$$

where A_n is the cross-section area normal to the solar radiation source and S is the solar flux in Watts/m^2. At 1 AU, the distance of the Earth, $S = 1361$ W/m^2; this value varies as $1/r^2$ with distance from the sun. At Jupiter ($\sim$5 AU), the solar flux is about 1/25th the value at Earth.

The same object will re-radiate heat as

$$Q_{out} = \epsilon A_{tot} \sigma T^4 \tag{8}$$

where A_{tot} is the total surface area, T is the object's temperature, and $\sigma = 5.67 \times 10^{-8}$ W/m^2/K^4 is the Stefan-Boltzmann constant (not to be confused with Boltzmann's constant which will come up later).

[1] These absorbance and emittance values are based on the typical solar spectrum; more rigorously, these coefficients are a function of wavelength.

In thermal equilibrium we must have $Q_{in} = Q_{out}$, so we can then easily find the equilibrium temperature of an object in space exposed to the sun:

$$T = \left(\frac{\alpha}{\epsilon}\right)^{1/4} \left(\frac{SA_n}{\sigma A_{tot}}\right)^{1/4} \tag{9}$$

We can see from this equation that the temperature essentially depends on two ratios: the ratio of the cross section area to the total surface area A_n/A_{tot}, and the ratio of the absorbance to the emittance. In areas where a spacecraft is likely to get hot, it pays to use a material (or coating) with a low α/ϵ; in areas that may get cold, the opposite is true. In this way a spacecraft can be designed to carefully use materials and coatings to optimize the temperature of spacecraft components.

Note that this derivation excluded any heat created by the spacecraft itself. Assuming it is burning some power in order to do something cool and useful, this excess heat must also be radiated. In that case the heat generated Q_{gen} must be added to equation (8), and we can derive the temperature as:

$$T = \left(\frac{\alpha SA_n + Q_{gen}}{\epsilon\sigma A_{tot}}\right)^{1/4} \tag{10}$$

In this case, the temperature is not as simple as the ratios mentioned earlier, but the same rules hold true: to dissipate heat, use a material or coating with low α/ϵ.

4.2 Radiation

The Sun is constantly emitting UV and X-ray radiation that can have damaging effects on spacecraft and humans in space. At lower altitudes, atmospheric shielding protects us from most UV damage. However, the polar regions are more vulnerable to radiation due to "open" magnetic field lines allowing particle access to lower altitudes. High particle flux dangers increase in this area when solar flares cause SEP events at Earth. Geomagnetic storms caused by CMEs or CIRs can reconfigure the Earth's magnetic field and allow damaging radiation access to latitudes even lower than the typical polar region. Another region with heightened particle flux levels is called the South Atlantic Anomaly, and we will discuss in later modules. Since we will cover the atmosphere, ionosphere, magnetosphere, and radiation belts in later modules, the rest of this section will try to focus on direct solar radiation and SEP events.

UV radiation from the Sun is energetic enough to break molecular bonds and degrade material properties of spacecraft and their electronics. Table 2.10 in Tribble (p. 50) shows many organic chemical bonds that can be broken by a single photon of UV light. For example, the Betacloth liner used in the bay of the space shuttle darkens visibly due to UV exposure when in orbit. The darkening of an exterior surface on a spacecraft will cause it to absorb more heat from the Sun. Exterior materials must be carefully selected so that UV degradation does not push temperatures higher than on board instruments are able to operate.

Solar radiation poses a threat to manned space flight. At a high enough energy, particles will pass through spacecraft and humans without any damaging effects. However, if the particles are stopped by human tissue, they can cause serious damage and ionize atoms within the body. Nearby atoms can become excited and produce more ionization in the body as they return to lower energy states.

We'll discuss radiation effects in more detail when we cover the Earth's radiation environment in Module 5.

4.3 Solar Panels

Not all effects of the sun on technology are bad! The sun provides the primary source of power for modern spacecraft. We know that the TSI doesn't vary much, so we have a reliable power source of 1361 W/m^2 at 1 AU. Solar panel area, efficiency, and distance from the sun are all important factors in determining the power that a panel will provide to a spacecraft.

5 Mission: Parker Solar Probe

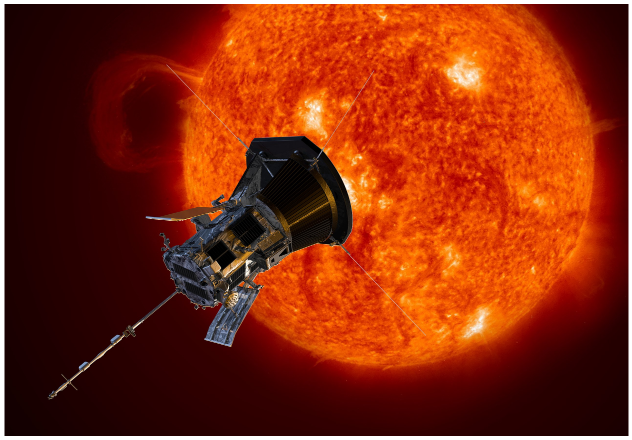

Figure 21: Parker Solar Probe. Courtesy NASA/Johns Hopkins APL/Steve Gribben.

NASA announced the Solar Probe Plus mission in 2009 after a competition with multiple proposals. The mission was developed over the next eight years, and before launch was renamed "Parker Solar Probe" in honor of Eugene Parker. It is the first NASA mission to be named after a living scientist. The mission is sometimes referred to in shorthand as Parker or PSP.

What sets Parker apart from previous solar missions is that it is the first mission to fly directly through the solar corona, and it has already flown closer to the sun than any previous man-made object (with a closest approach inside 10 solar radii). It's primary goal is to figure out how the corona is heated to millions of Kelvins, and determine how the solar wind is accelerated to hundreds of km/s. We observe these temperatures and speeds *remotely*, but Parker will make *in-situ* measurements.

Parker launched on August 12, 2018. During its first week in space it deployed its high-gain antenna to communicate back to Earth, its magnetometer boom, and its electric field antennas. Instrument testing began in early September, and its first science observations were transmitted back to Earth in December 2018. Parker Solar Probe completed its 24th planned perihelion pass on June 19, 2025. It is on track for more passes in its current orbit, with the next one planned for September 15, 2025. The plan is to continue in this orbit, but requiring adjustment to maintain attitude so that its transmitters point at Earth. Eventually, its thrusters will run out of fuel, and full functioning will not then be possible.

For more information on Parker: https://en.wikipedia.org/wiki/Parker_Solar_Probe.

Problems

1. **(10) Radiation from the Corona:** i) If the temperature in a region of the corona is 2 MK (two million Kelvins), what is the characteristic wavelength emitted by this region? What type of electromagnetic wave is this? ii) Repeat for a cooler region of the photosphere at 5,778 K.

2. **(10) Thermal escape:** What is the thermal velocity of a proton in the outer corona where the temperature is 2 MK? (Hint: look up "thermal velocity" on wikipedia. Consider three dimensions.) Is this fast enough to escape the gravitational field of the sun? (Hint 2: look up "escape velocity".) Are coronal protons hot enough to escape from the sun's gravity?

3. **(10) Solar Wind:** i) The quiescent solar wind density at Earth is about 5 ions/cm^3, moving at a velocity of 400 km/s. If most of these ions are protons, what is the total mass ejected from the sun per second? ii) Given these parameters, what is the solar wind particle flux (ions/cm^2/sec) at the orbit of Jupiter?

4. **(10) Relativistic Protons:** Assuming that most solar protons are accelerated in the lower corona ($\sim$2 R_S), create a plot that shows distance traveled in AU vs time for 1, 10, 100, and 500 MeV protons. How long in hours does it take for each of these energies to reach Earth? (refer to USW (Understanding Space Weather and ...) Section 2.1.4).

5. **(15) Heliopause:** The Heliopause occurs where the pressure of the interstellar medium balances the pressure of the solar wind. If the interstellar pressure is $\sim 1.3 \times 10^{-13}$ Pa, calculate the distance from the sun to the heliopause in AU. Again assume the solar wind density is 5 ions/cm^3 at 1 AU.

6. **(15) Spacecraft Temperature:** Assume that aluminum has a solar absorptivity of 0.15 and an emissivity of 0.05. i) Imagine a spherical probe orbiting the Earth; what is its equilibrium temperature when it is in sunlight? ii) What about the same probe orbiting Jupiter? iii) and iv) Repeat, but now assume the spacecraft is coated with GSFC White Paint NS-74, which has a solar absorptivity of 0.17 and emissivity of 0.92. What do you think this paint is designed for??

7. **(15) Solar Outputs:** Use Figure 22 to answer the following questions:

   (a) What is the minimum radiative energy flux of an M-Class flare?

   (b) Consider the flare at the beginning of March 07, 2012. Assume the peak energy flux on this plot is 5×10^{-4} W/m^2 (it's a bit hard to see because of the text). Determine the corresponding energy flux at the emission source location of 1.5 solar radii.

   (c) The flares with the sharp rise and multi-hour decay (long duration events, or LDEs) are the ones most likely to have an associated Coronal Mass Ejection (CME). About how many CMEs were **likely** launched into the heliosphere in March 2012 according to this chart?

   (d) Comment on the relationship between the X-ray flares (top panel) and the flux of solar protons (second panel).

   (e) Comment on the relationship between cosmic rays (the third panel) and the two panels above it. (We'll ignore the magnetic field signatures until the next module.)

8. **(15) Parker Solar Probe:** Parker Solar Probe has five "investigations": FIELDS, ISIS, WISPR, SWEAP, and HeliOSPP. Each of these investigations includes multiple instruments. Do some research and pick your favorite instrument (not the entire investigation). In a couple of paragraphs (or in bullet points if you prefer), explain:

   - What does this instrument do?

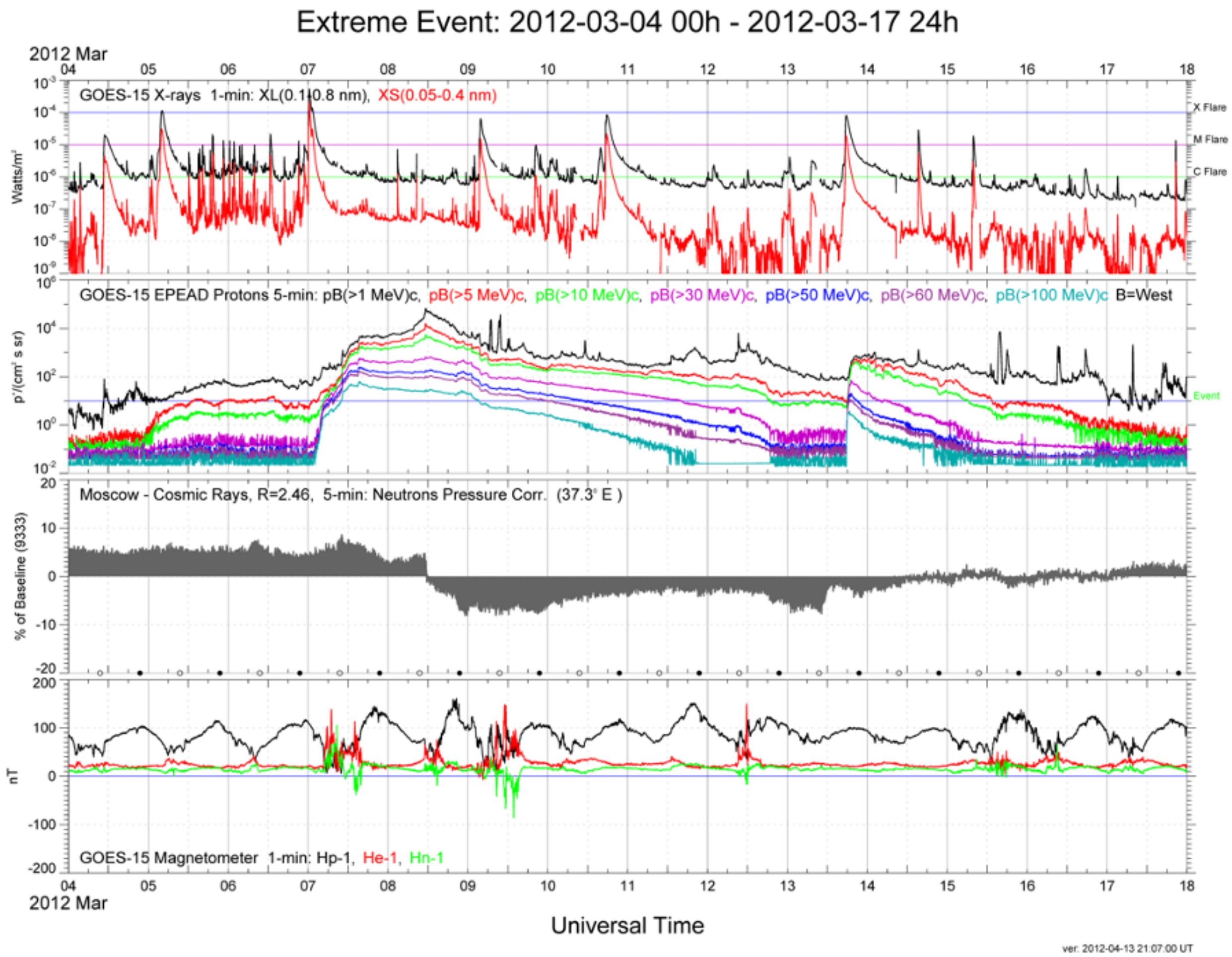

Figure 22: Outputs from the sun in March 2012. From top to bottom: i) Radiative energy flux measured at 1 AU in two wavelength bands; ii) energetic protons measured at 1 AU in a range of energy bands; iii) cosmic ray flux variation measured on the ground near Moscow; and iv) three-axis magnetic field intensity measured by GOES at 1 AU.

- What is the instrument SWAP (size, weight, and power)?
- In basic terms, how does it work?
- How does it contribute to the science goals of its parent investigation?